METHIOPEPTIDE®

Cell respiration recovery
Skin energizer
Anti-aging: « 2nd wind » effect
Anti-stress: long lasting protection
Photo-protection

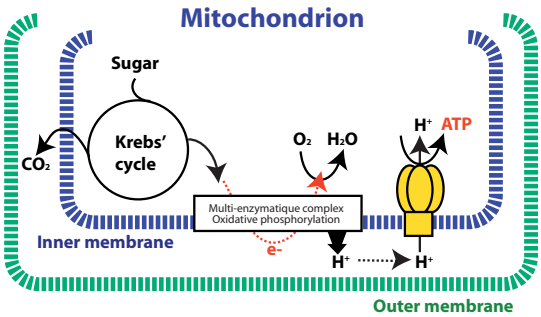
EXSYMOL
MONACO

METHIOPEPTIDE®

AGING CELL RESPIRATION

Oxygen (O_2) is used by our organism to convert energetic molecules (sugars, aminoacids and lipids) into adenosine triphosphate (ATP), a versatile biochemical form of energy used to fuel metabolic processes. Cell respiration (O_2 consumption), occurs within mitochondria. These small cell sub-units can be considered as the cell's « powerhouses ».

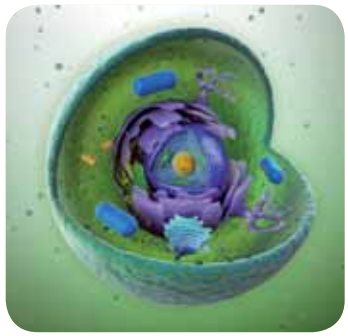
Cell respiration (oxidative phosphorylation) involves a multi-enzymatic complex located in mitochondria's inner membrane.



Oxygen is the final acceptor molecule of electrons transferred within this complex, also called "respiratory chain". Electron transfer is coupled with proton exclusion from mitochondria inner compartment. Proton gradient enables ATP synthesis.

Mitochondria are vulnerable power plants

Many free radicals derived from oxygen (ROS) are formed within mitochondria either because of sunrays (UV and IR) or because of electron leakage from the respiratory chain.



The respiratory chain is damaged by ROS and this result in an increased electron leakaged that also form ROS. All these ROS participate to skin premature aging.

Cell respiration declines with aging

With aging, the accumulation of non-repaired damages, due to ROS, results in a gradual impairment of the respiratory function (less ATP is produced). As a consequence, more and more electron leakage occurs, and ROS formation within the mitochondria is increased.

This respiration decline initiates a vicious circle: the energy production decrease weakens defense and repairing systems, which results in accelerated damages accumulation.

EXSYMOL 's SOLUTION:
a biobetter protector designed for recovery of the respiratory function

- **BIOMIMETIC**: METHIOPEPTIDE® contains methionine, an antioxidant aminoacid present at high levels in mitochondrial proteins. METHIOPEPTIDE® doesn't affect the oxidative phosphorylation process and is also recognized by MSR (Methionine Sulfoxide Reductase), an enzyme responsible for methionine recycling.

- **BIOBETTER**: METHIOPEPTIDE® is much more effective than methionine.

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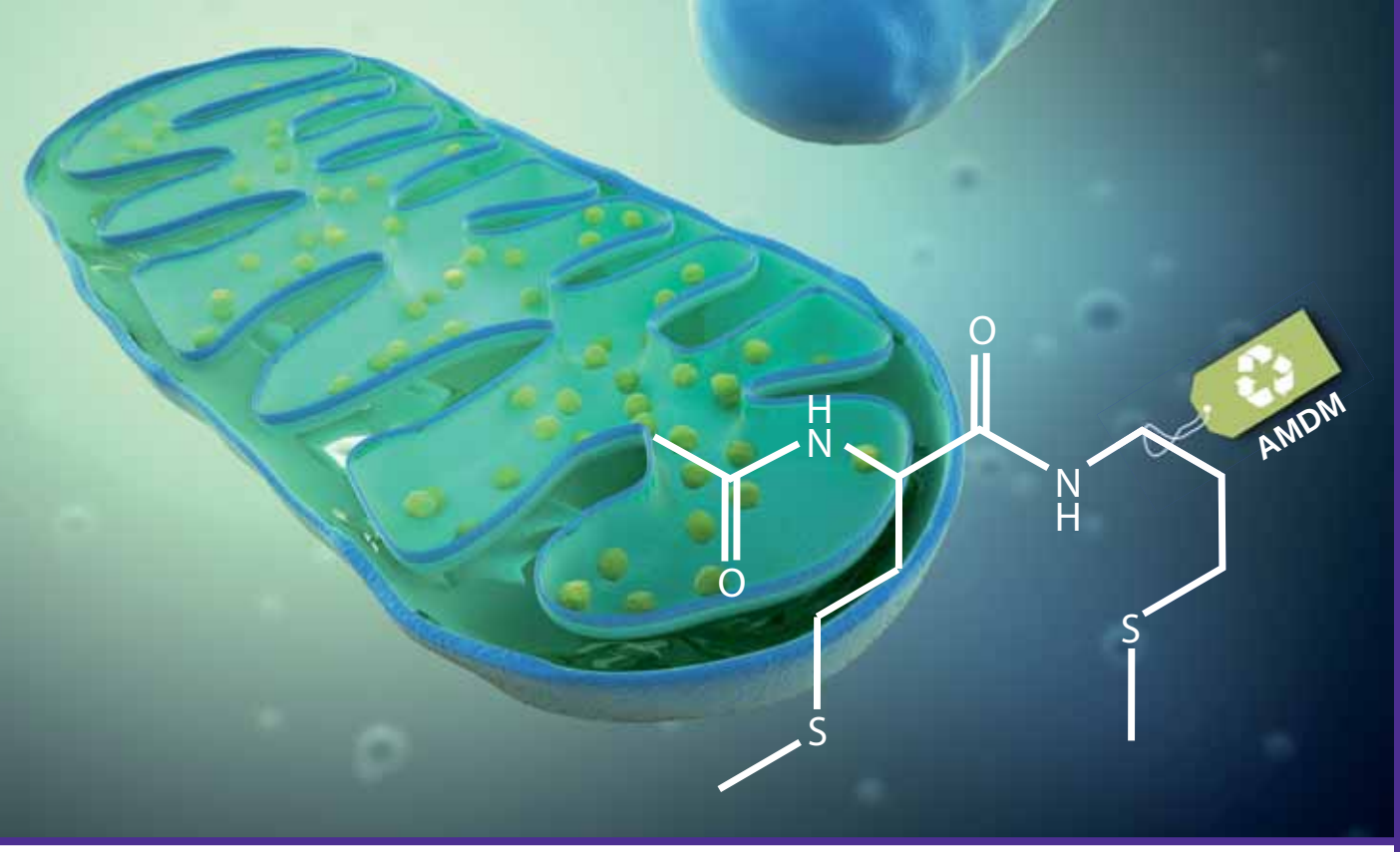

METHIOPEPTIDE®

INCI name : ACETYL METHIONYL DECARBOXYMETHIONINE

METHIOPEPTIDE® is a preservative-free 15% hydroglycolic solution of AMDM (Acetyl methionyl decarboxymethionine), a « biobetter » peptide derived from dimethionine.

- Inspired by skin natural defense
- Safe (no toxic end products)
- Recycled by skin enzymes (MSR)
- More efficient than natural methionine
- Bioavailable

SKIN BENEFITS

- Respiratory function recovery
- Mitochondria protection
- DNA protection
- Cell protection
- Photoprotection
- Intra-mitochondrial ROS scavenging

COSMETIC APPLICATION

- Anti-aging
- Skin energizer: cell respiration protection
- Energy recycling
- Photo-protection
- Cells 2nd wind - Metabolic recovery
  - 1-Enhanced natural skin defense (epidermal survival)
  - 2-Optimized cell longevity
- Improved Sun protection

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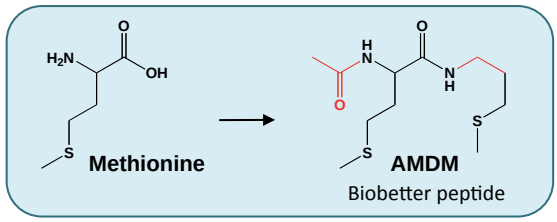
METHIOPEPTIDE®

BIO-BETTER technology

Get inspired by nature and exceed its potential

Methionine is a versatile antioxidant aminoacid that participates to proteins natural defense. It is known as the "last chance defense system", since it scavenges ROS when all other systems failed (antioxidant enzymes and antioxidant vitamins). Mitochondrial proteins, that are exposed to an important oxidative stress, are rich in methionine.



METHIOPEPTIDE® was designed for performance. Structural modifications improved its bioavailability (methionine penetration in skin and cells is low) and its reactivity towards ROS.

METHIOPEPTIDE® is a biobetter

METHIOPEPTIDE® is an improved methionine that is naturally present in all cells. It has a better resistance to enzymatic degradation (1), better cell penetration (2), a better skin penetration (3), and better scavenging abilities (4).

1 *Resistance to hydrolytic enzymes*

AMDM structural characteristics avoid Methiopeptide degradation in the superficial layers of the skin

2 *Penetration into skin cells (horizontal bio-availability)*

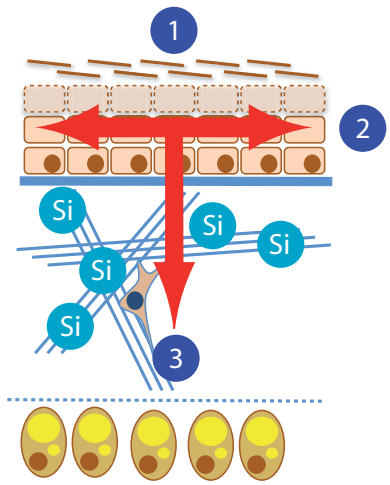
	λ
Ac Methionine	0.67
AMDM	2.03

3 *Skin penetration (vertical bio-availability)*

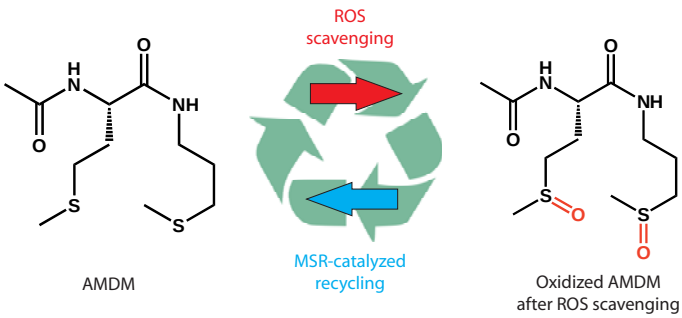
	Log D (pH7)
Ac Methionine	-4.65
AMDM	0.6

4 *Scavenging of hydroxyl radicals (OH°)*

	OH° scavenged at 30 mM
Ac Methionine	44%
AMDM	83%



Regeneration by MSR, an enzymatique recycling system



ROS-mediated oxidation of methionine is reversible. Indeed, oxidized methionines within proteins are reduced into native methionines by the action of Methionine Sulfoxide Reductase (MSR), an enzyme naturally present in the skin.

Thanks to AMDM close resemblance with methionine, the oxidized forms of AMDM can be recognized and regenerated by MSR. Hence, METHIOPEPTIDE® can provide the skin with a long lasting protection.

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OPTIMIZED CELL RESPIRATION

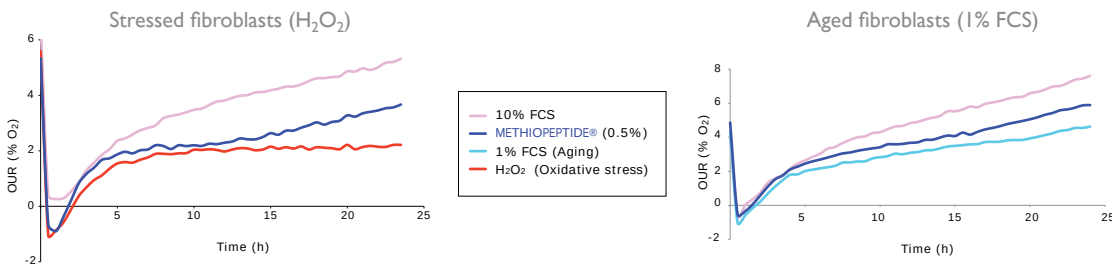
METHIOPEPTIDE®

Oxygen is essential for life, and almost all of it is used to produce energy (ATP synthesis in the respiratory chain). As a consequence, oxygen consumption monitoring (oxymetry) is an accurate method to determine cell health, and cell response to stress (ROS, pollutants, UV).

In vitro oxymetry test

With age, or because of a stress, the cell respiration process is affected. As a result, fewer energy (ATP) will be produced.

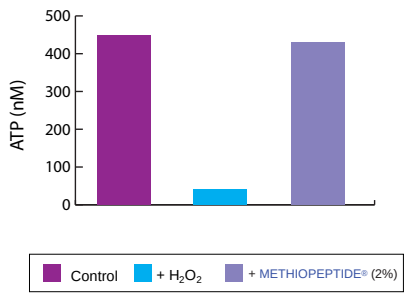
METHIOPEPTIDE® (0.5%) is capable of protecting the cell respiration from aging and from the noxious effects of an oxidative stress.



A protected energy synthesis

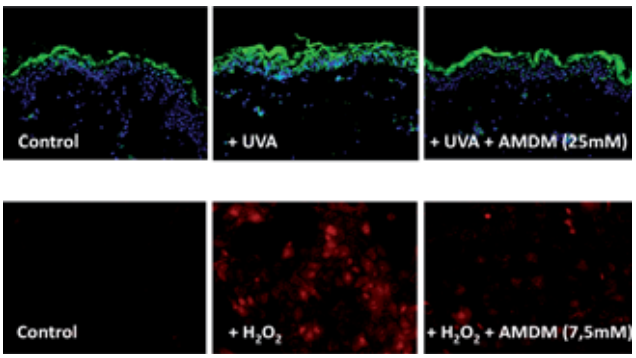
In the case of an H_2O_2-induced oxydative stress, cell oxygen consumption decreases leading to a decrease of ATP production.

METHIOPEPTIDE® (2%) managed to maintain an optimal energy level.



Mitochondria protection

Mitochondrie are responsible fir supplying the cells with energy. This (organels) are vulnerable to stress. It is therefore of primary importance to protect them from harm.



Exposure to UV rays leads to an intra-mitochondrial ROS overproduction (green). METHIOPEPTIDE® (5%) is capable of reducing this UV-induced ROS production.

When mitochondria are exposed to this oxidative stress they are damaged (damaged mitochondria marked in red) and/or destroyed. METHIOPEPTIDE® (1.5%) is able to protect mitochondria from oxidative stress.

By protecting mitochondria, METHIOPEPTIDE contributes to an optimal energy supply for the skin.

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METHIOPEPTIDE®

SKIN 2nd WIND

Skin benefits of a rejuvenated cell respiration

With an energetic increase of cell metabolism, skin will benefit of high level of protection and health.

Global protection

METHIOPEPTIDE® is capable of protecting skin cells at all levels:

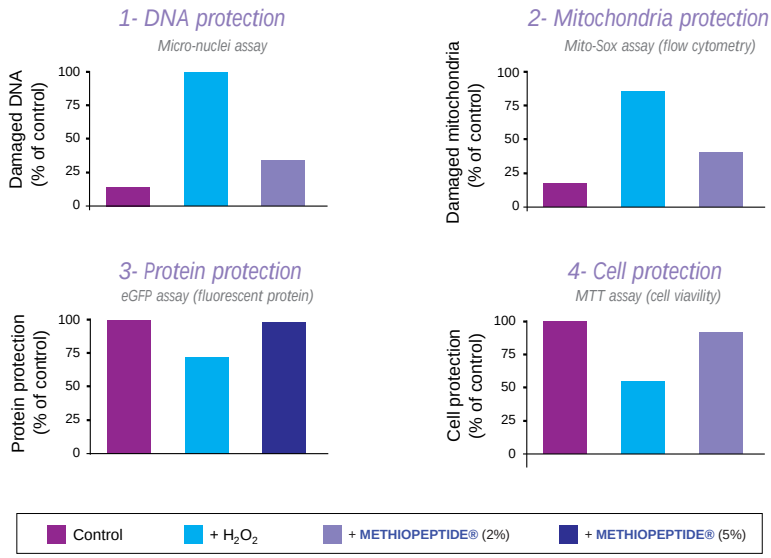
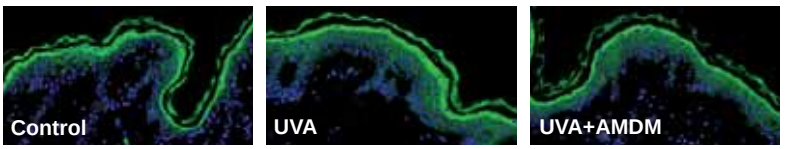
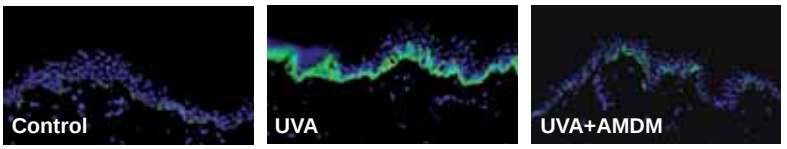


Photo-protection

Because of sunrays exposure, a few reaction in the skin that are responsible for the weakening of natural defenses may happen.



METHIOPEPTIDE® is able to prevent the apparition of oxidized tyrosine (marked in green) hence to maintain optimal defense mechanisms for the skin.



Because the skin cells are better protect, the number of dying cells (marked in green) is substantially decreased in the case of a treatment with METHIOPEPTIDE®.

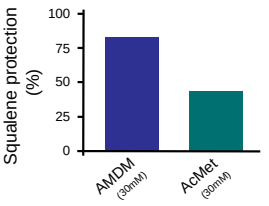
Sun care – special feature

Many studies have shown that, in order to ease the free radical quenching, there is an overproduction of MSR, the enzyme responsible for the recycling of methionine sulfoxide back into methionine (ref).

Since MSR is also able to recycle AMDM, an increased quantity of MSR also means a higher quantity of AMDM available for the skin to use as a photoprotector.

Prevention of greasy skin - Anti-comedogenic

Squalenes are the first lipids on skin surface that suffer from radiation-induced oxidation. Once oxidized, they become toxic, create an inflammation and form comedones (blackheads). AMDM protects these squalenes and therefore the skin quality.



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TECHNICAL CHARACTERISTICS

METHIOPEPTIDE®

ANALYTICAL COMPOSITION

AMDM.....15%
METHYLPROPANEDIOL.....40%
WATER.....45%

PHYSICO-CHEMICAL CRITERIA

Limpid and colorless liquid.
pH ≈ 5.5.
Miscible with water, alcohol and glycol.

PRESERVATIVE

No preservatives.

TOLERANCE STUDY

METHIOPEPTIDE® is perfectly tolerated. Tolerance and toxicity studies were performed using both *in vitro* (cell culture and reconstructed epidermis) and *in vivo* (human volunteers) methods.

FORMULATION

Advised doses: 2%.
Store at room temperature.

AVAILABILITY

Available in 1, 5, 30 and 60 kg drums.

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